

CLIMATE CHANGE SCIENCE GREENHOUSE EFFECT

GREENHOUSE-EFFECT FIREWALL → ENERGY-BALANCE RAIL → EMISSION-TO-CONCENTRATION LEDGER → FORCING-FEEDBACK FORK → WATER AND PARTICLE MATRIX → GAS COMPARISON GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 GREENHOUSE-EFFECT FIREWALL

GREENHOUSE-EFFECT FIREWALL NATURAL EFFECT ABSORBS AND RE-EMITS OUTGOING LONGWAVE RADIATION MAINTAINS A HABITABLE ENERGY BALANCE HUMAN ENHANCEMENT RAISES CONCENTRATIONS AND NET FORCING CLIMATE RESPONSE WARMING PLUS WIDER SYSTEM CHANGES

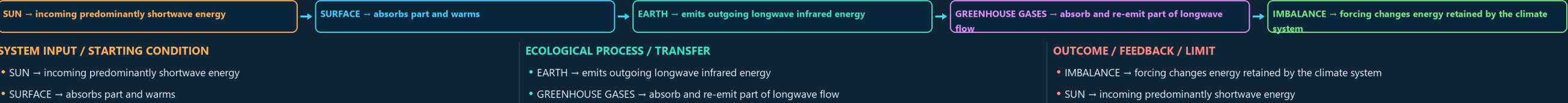
ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation	HUMAN ENHANCEMENT → raises concentrations and net forcing	EXAM RULE → preserve the natural-versus-enhanced distinction
FUNCTION → maintains a habitable energy balance	CLIMATE RESPONSE → warming plus wider system changes	NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation
ECOLOGICAL UNIT / PARAMETER - NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation - FUNCTION → maintains a habitable energy balance	DECISIVE DISTINCTION - HUMAN ENHANCEMENT → raises concentrations and net forcing - CLIMATE RESPONSE → warming plus wider system changes	SCALE / STATUS / EXCEPTION - EXAM RULE → preserve the natural-versus-enhanced distinction - NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLIMATE RESPONSE → warming plus wider system changes.

01

STAGE 01 ENERGY-BALANCE RAIL

ENERGY-BALANCE RAIL INCOMING PREDOMINANTLY SHORTWAVE ENERGY ABSORBS PART AND WARMS EMITS OUTGOING LONGWAVE INFRARED ENERGY GREENHOUSE GASES ABSORB AND RE-EMIT PART OF LONGWAVE FLOW FORCING CHANGES ENERGY RETAINED BY THE CLIMATE SYSTEM

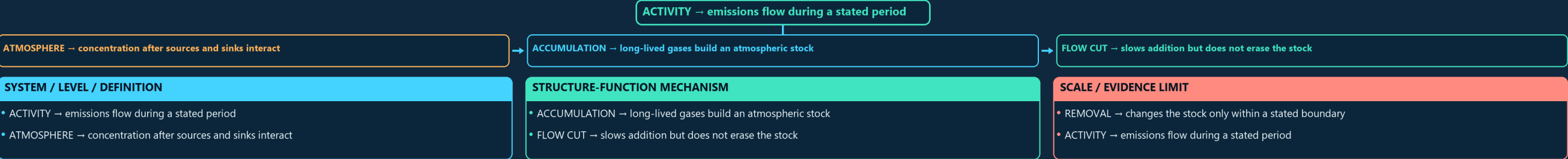


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IMBALANCE → forcing changes energy retained by the climate system.

02

STAGE 02 EMISSION-TO-CONCENTRATION LEDGER

EMISSION-TO-CONCENTRATION LEDGER EMISSIONS FLOW DURING A STATED PERIOD CONCENTRATION AFTER SOURCES AND SINKS INTERACT LONG-LIVED GASES BUILD AN ATMOSPHERIC STOCK FLOW CUT SLOWS ADDITION BUT DOES NOT ERASE THE STOCK CHANGES THE STOCK ONLY WITHIN A STATED BOUNDARY

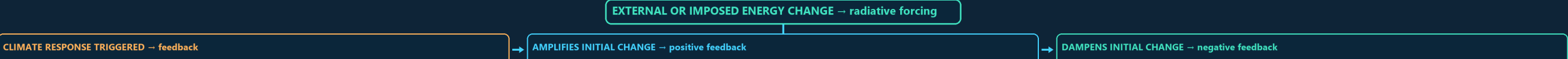


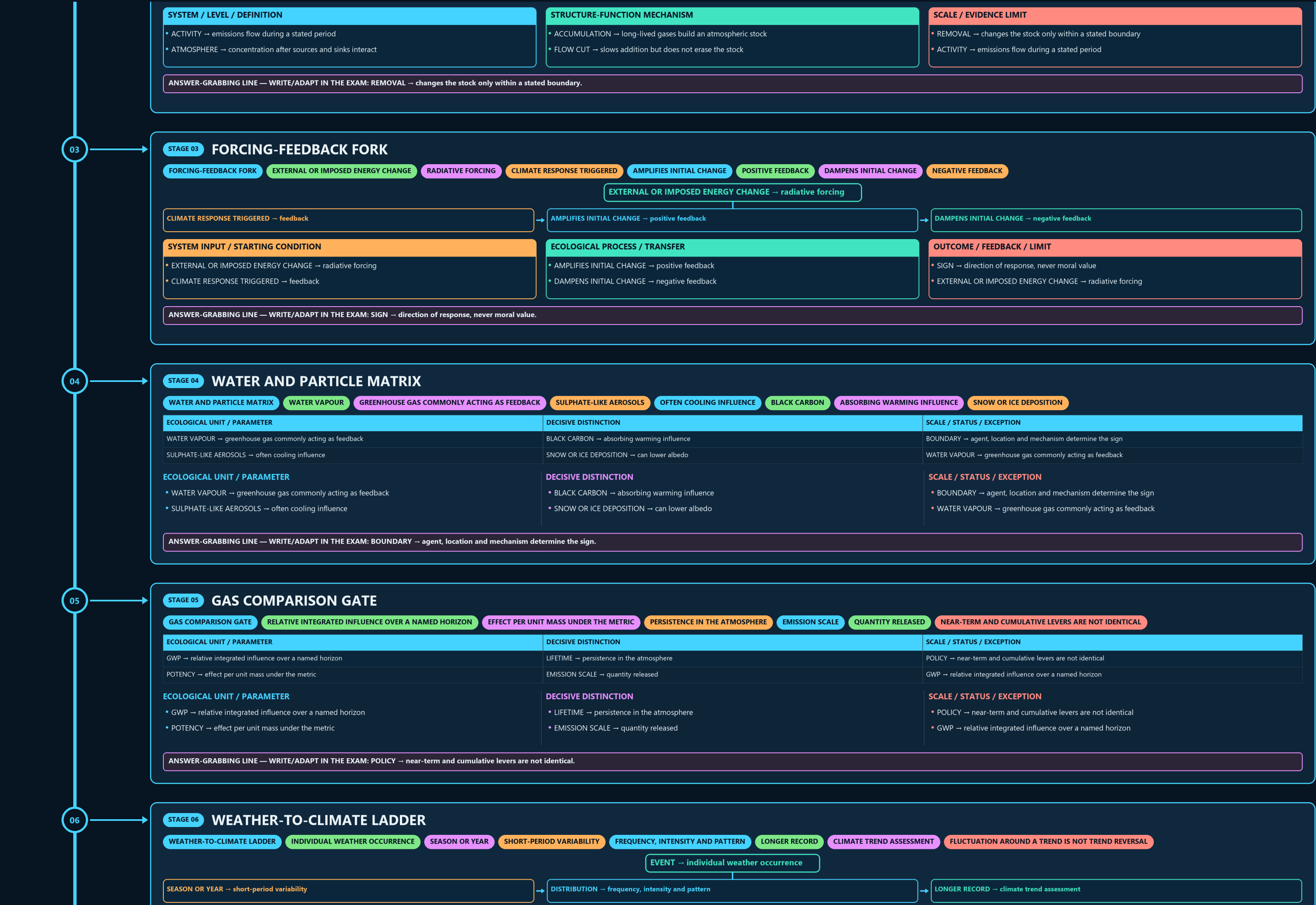
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REMOVAL → changes the stock only within a stated boundary.

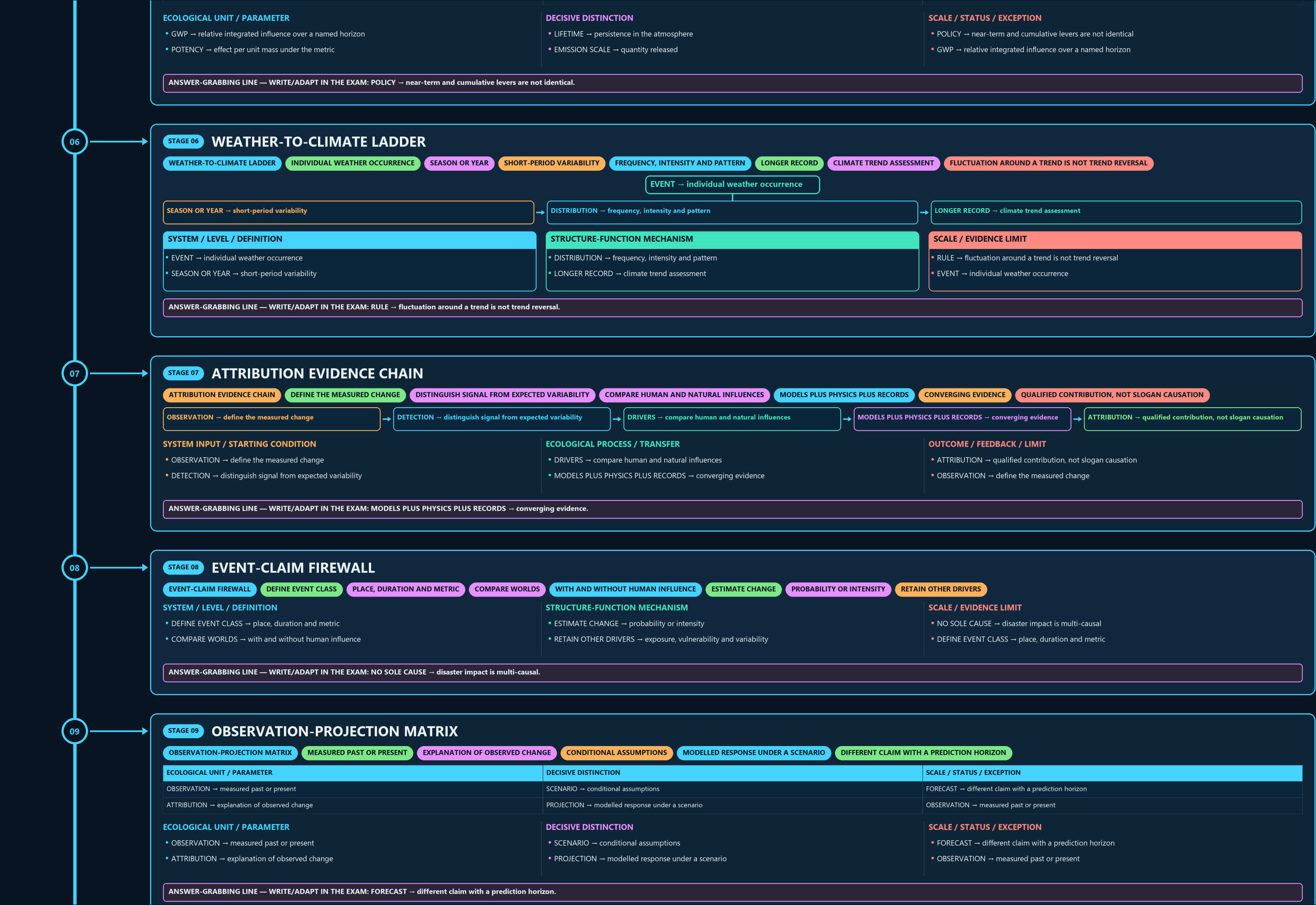
03

STAGE 03 FORCING-FEEDBACK FORK

FORCING-FEEDBACK FORK EXTERNAL OR IMPOSED ENERGY CHANGE RADIATIVE FORCING CLIMATE RESPONSE TRIGGERED AMPLIFIES INITIAL CHANGE POSITIVE FEEDBACK DAMPENS INITIAL CHANGE NEGATIVE FEEDBACK







09

STAGE 09 OBSERVATION-PROJECTION MATRIX

OBSERVATION-PROJECTION MATRIX

MEASURED PAST OR PRESENT

EXPLANATION OF OBSERVED CHANGE

CONDITIONAL ASSUMPTIONS

MODELLED RESPONSE UNDER A SCENARIO

DIFFERENT CLAIM WITH A PREDICTION HORIZON

ECOLOGICAL UNIT / PARAMETER

OBSERVATION → measured past or present

ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION

SCENARIO → conditional assumptions

PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION

FORECAST → different claim with a prediction horizon

OBSERVATION → measured past or present

ECOLOGICAL UNIT / PARAMETER

• OBSERVATION → measured past or present

• ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION

• SCENARIO → conditional assumptions

• PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION

• FORECAST → different claim with a prediction horizon

• OBSERVATION → measured past or present

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FORECAST → different claim with a prediction horizon.

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STAGE 10 CLIMATE-RESPONSE FORK

CLIMATE-RESPONSE FORK

SOURCE REDUCTION

SINK ENHANCEMENT

MITIGATION WITH ACCOUNTING BOUNDARY

EXPOSURE OR VULNERABILITY REDUCTION

RESIDUAL HARM

SEPARATE LOSS-AND-DAMAGE QUESTION

MITIGATION AND ADAPTATION ARE COMPLEMENTARY

SOURCE REDUCTION → mitigation

SINK ENHANCEMENT → mitigation with accounting boundary

EXPOSURE OR VULNERABILITY REDUCTION → adaptation

RESIDUAL HARM → separate loss-and-damage question

SYSTEM / LEVEL / DEFINITION

• SOURCE REDUCTION → mitigation

• SINK ENHANCEMENT → mitigation with accounting boundary

STRUCTURE-FUNCTION MECHANISM

• EXPOSURE OR VULNERABILITY REDUCTION → adaptation

• RESIDUAL HARM → separate loss-and-damage question

SCALE / EVIDENCE LIMIT

• PORTFOLIO → mitigation and adaptation are complementary

• SOURCE REDUCTION → mitigation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESIDUAL HARM → separate loss-and-damage question; PORTFOLIO → mitigation and adaptation are complementary.

11

STAGE 11 SCIENCE ANSWER SPINE

SCIENCE ANSWER SPINE

NATURAL EFFECT, HUMAN ENHANCEMENT AND ENERGY FLOW

EMISSIONS, CONCENTRATIONS, STOCK AND NET BALANCE

FORCING, FEEDBACK AND GAS-SPECIFIC PROPERTIES

TREND, DETECTION, ATTRIBUTION AND PROJECTION

SCALE, SCENARIO, CONFIDENCE AND NO INVENTED FIGURES

DEFINE → natural effect, human enhancement and energy flow

ACCOUNT → emissions, concentrations, stock and net balance

EXPLAIN → forcing, feedback and gas-specific properties

EVALUATE → trend, detection, attribution and projection

QUALIFY → scale, scenario, confidence and no invented figures

SYSTEM / LEVEL / DEFINITION

• DEFINE → natural effect, human enhancement and energy flow

• ACCOUNT → emissions, concentrations, stock and net balance

STRUCTURE-FUNCTION MECHANISM

• EXPLAIN → forcing, feedback and gas-specific properties

• EVALUATE → trend, detection, attribution and projection

SCALE / EVIDENCE LIMIT

• QUALIFY → scale, scenario, confidence and no invented figures

• DEFINE → natural effect, human enhancement and energy flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → scale, scenario, confidence and no invented figures.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S NATIONAL EMISSIONS PROFILE (AS REPORTED IN ITS PERIODIC

ANALYTICAL POINT

INDIA'S PER-CAPITA AND HISTORICAL CUMULATIVE EMISSIONS REMAIN

GLOBAL EMISSIONS AND REMAINING-CARBON-BUDGET FIGURES ARE PERIODICALLY UPDATED BY

ATTRIBUTION-SCIENCE CONFIDENCE LEVELS VARY MEANINGFULLY BY EVENT TYPE (HIGH

INDIA-SPECIFIC SECTORAL EMISSIONS DATA (FROM NATIONAL COMMUNICATIONS/BURS) IS REPORTED

IPCC AR6 ATTRIBUTION SCIENCE HAS SUBSTANTIALLY STRENGTHENED CONFIDENCE IN

OPTIONAL ECOLOGICAL PROCESS DEPTH

• CAUTION: India's national emissions profile (as reported in its periodic UNFCCC national

• CAUTION: Analytical point: India's per-capita and historical cumulative emissions remain

• CAUTION: Global emissions and remaining-carbon-budget figures are periodically updated by IPCC

• CAUTION: Attribution-science confidence levels vary meaningfully by event type (high confidence

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: India-specific sectoral emissions data (from national communications/BURS) is reported

• IPCC AR6 attribution science has substantially strengthened confidence in linking

• Aerosols exert a net cooling (negative radiative forcing) effect that partially masks

• The remaining carbon budget concept reframes climate policy around a finite, depleting

COMPARATIVE RESTORATION NUANCE

• India's methane and nitrous oxide emissions are disproportionately driven by

• NOT: All extreme weather events can be attributed to climate change with equally high

• NOT: Reducing air pollution (aerosols) has no bearing on the rate of global warming. - It

• NOT: A distant net-zero target date alone is sufficient to remain within a fixed carbon

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.